SG No. 101 C_{4v}^3 $P4_2cm$ [tetragonal]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `2a`, site symmetry: `2.mm`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, 0, z]$	`[1,-2,-7,8]`
2	$[-X, X, 0]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[3,-4,-5,6]`

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,7,8]`
2	$[0, 0, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[3,4,5,6]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, 0, z]$	`[1,8]`
2	$[-X, -X, Z]$	$[0, 0, z]$	`[2,7]`
3	$[-X, X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[3,6]`
4	$[X, -X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[4,5]`

Table 4: Wyckoff bond: `4d@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[5,-6]`
4	$[-Y, -X, 0]$	$[0, 0, z]$	`[7,-8]`

Table 5: Wyckoff bond: `8e@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$[0, 0, z]$	`[7]`
8	$[Y, X, Z]$	$[0, 0, z]$	`[8]`

* Wyckoff site: `2b`, site symmetry: `2.mm`

Table 6: Wyckoff bond: `2a@2b`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,-2,-7,8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[3,-4,-5,6]`

Table 7: Wyckoff bond: `2b@2b`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[3,4,5,6]`

Table 8: Wyckoff bond: `4c@2b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,8]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2,7]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[3,6]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[4,5]`

Table 9: Wyckoff bond: `4d@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[5,-6]`
4	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[7,-8]`

Table 10: Wyckoff bond: `8e@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[8]`

* Wyckoff site: `4c`, site symmetry: `2..`

Table 11: Wyckoff bond: `4a@4c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[5,-6]`
4	$[-Y, -X, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[7,-8]`

Table 12: Wyckoff bond: `4b@4c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[3,4]`
3	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[5,6]`
4	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[7,8]`

Table 13: Wyckoff bond: `8c@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[8]`

* Wyckoff site: `4d`, site symmetry: `..m`

Table 14: Wyckoff bond: `4a@4d`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$[x, x, z]$	`[1,8]`
2	$[-X, -X, Z]$	$[-x, -x, z]$	`[2,7]`
3	$[-X, X, Z]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[3,6]`
4	$[X, -X, Z]$	$\left[x, -x, z+\frac{1}{2}\right]$	`[4,5]`

Table 15: Wyckoff bond: `4b@4d`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	`[1,-8]`
2	$[-X, X, 0]$	$[-x, -x, z]$	`[2,-7]`
3	$[X, X, 0]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[3,-6]`
4	$[-X, -X, 0]$	$\left[x, -x, z+\frac{1}{2}\right]$	`[4,-5]`

Table 16: Wyckoff bond: `8c@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, -x, z+\frac{1}{2}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[x, -x, z+\frac{1}{2}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$[-x, -x, z]$	`[7]`
8	$[Y, X, Z]$	$[x, x, z]$	`[8]`

* Wyckoff site: `8e`, site symmetry: `1`

Table 17: Wyckoff bond: `8a@8e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[-y, x, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y, -x, z+\frac{1}{2}\right]$	`[4]`
5	$[X, -Y, Z]$	$\left[x, -y, z+\frac{1}{2}\right]$	`[5]`
6	$[-X, Y, Z]$	$\left[-x, y, z+\frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$[-y, -x, z]$	`[7]`
8	$[Y, X, Z]$	$[y, x, z]$	`[8]`